

# Probing Quantum Nonlocality with Spacetime-Resolved $H \rightarrow \tau^+ \tau^-$ Decays

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We show that a future  $e^+e^-$  Higgs factory can perform a spacetime-resolved test of quantum nonlocality in Higgs boson decays. In  $e^+e^- \rightarrow ZH \rightarrow \mu^+\mu^-\tau^+\tau^-$  events at  $\sqrt{s} = 240$  GeV, the tau decay vertices are reconstructed with the impact-parameter method, giving event-by-event access to the spacetime interval between the two decays. Because the Higgs is a scalar, the  $\tau^+\tau^-$  pair is produced in the maximally entangled triplet  $(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)/\sqrt{2}$ , with spin correlation matrix  $C = \text{diag}(+1, +1, -1)$  exactly, for any tau velocity. We construct unbiased estimators of the CHSH score and of a concurrence witness, derive the exact small-sample noise bias of the Horodecki parameter, and build a Neyman–Pearson-optimal, permutation-calibrated test of the hypothesis that the correlations are carried by a signal of finite speed  $v_\psi$ . The causal-speed distribution is shown to follow an exact Pareto law,  $P(v_{\text{sig}} > v) = \beta/v$ , yielding the closed-form exclusion reach  $v_\psi^{95} \simeq N\beta/(4z_{95}^2)$ . In a simulated sample of 90 fully reconstructed  $\pi^+\pi^-$  events—all with spacelike-separated decay vertices—entanglement is certified at  $3.8\sigma$  and finite-speed causal models are excluded up to  $v_\psi \approx 3c$ , in agreement with the analytic reach. Scaling to the full ILC program, the exclusion extends beyond  $10^2c$ , providing the first test of quantum nonlocality in a Yukawa-mediated process with massive fermions.

The decay  $H \rightarrow \tau^+\tau^-$  produces a maximally entangled pair of massive fermions. Because the Higgs boson is a spin-0 particle, the  $\tau^+\tau^-$  system is created in a pure, maximally entangled spin state, regardless of the tau velocity [19–21]. Measuring the spin correlations of this system tests quantum mechanics in the relativistic regime, with direct sensitivity to Bell nonlocality [1–3].

Recent landmark measurements at the LHC, building on the proposal of Ref. [15], have established quantum entanglement in top quark pairs [16, 17] and provide evidence for entangled  $H \rightarrow \tau^+\tau^-$  decays [18]. However, these measurements integrate over the spacetime structure of the decays: they demonstrate *that* the particles are entangled but do not probe *where* or *when* the entanglement is manifest. In photon-based Bell tests, the spacelike separation of measurements is enforced by the experimental design [4–6], and dedicated experiments bound the speed of any hypothetical superluminal influence [7]. At a collider, the tau leptons propagate freely and decay at spacetime positions governed by quantum mechanics; the spacelike separation is not imposed but *measured*.

There is also a historical motivation for testing quantum correlations specifically in weak-sector processes. The weak interaction has repeatedly overturned assumptions about fundamental symmetries: parity [9, 10], CP [11], and lepton-flavor conservation [12] were each found to be violated when tested. Quantum nonlocality has been verified extensively with photons [4–6] but never directly tested in an interaction mediated by the Higgs field. The Yukawa coupling  $y_\tau H \bar{\tau}\tau$  is the only known interaction that produces maximally entangled massive-fermion pairs from a scalar source. A direct test of nonlocality in this channel is therefore well motivated, inde-

pendent of theoretical prejudice.

In this Letter we demonstrate that a future  $e^+e^-$  Higgs factory such as the ILC [27] enables precisely this measurement. The enabling capabilities are: (i) the known beam four-momentum, which fixes the Higgs rest frame through the  $Z$  recoil; (ii) the impact-parameter method of Jeans [25], which reconstructs the tau decay vertex from charged-pion track geometry; and (iii) the micron-level vertex resolution of detectors such as the ILD [28]. We further develop the statistical framework required for a defensible claim: exact Standard-Model (SM) expectations, unbiased estimators with closed-form variances, an optimal likelihood-ratio (LR) test calibrated by permutation, and an analytic sensitivity law that is validated on simulation.

*Exact SM expectation.*—For the decay of a scalar to two spin- $\frac{1}{2}$  fermions, the spin density matrix of the  $\tau^+\tau^-$  pair is

$$\rho = \frac{1}{4} [\mathcal{K} \otimes \mathcal{K} + B_i^+ \sigma_i \otimes \mathcal{K} + B_j^- \mathcal{K} \otimes \sigma_j + C_{ij} \sigma_i \otimes \sigma_j], \quad (1)$$

with  $i, j \in \{n, r, k\}$  the orthonormal basis built from the  $\tau^-$  flight direction  $\hat{k}$  in the Higgs rest frame, the transverse beam component  $\hat{r}$ , and  $\hat{n} = \hat{k} \times \hat{r}$ . A CP-even scalar coupling  $\bar{\tau}\tau$  admits a single partial wave:  $J = 0$  with parity  $(-1)^{L+1} = +1$  forces  $L = S = 1$ , the  ${}^3P_0$  wave. Along the decay axis only  $m_L = 0$  survives, so the spin state is *exactly* the triplet  $|\psi^+\rangle = (|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)/\sqrt{2}$ , giving  $B_i^\pm = 0$  and

$$C_{\text{SM}} = \text{diag}(+1, +1, -1), \quad (2)$$

independent of the tau velocity  $\beta$ . More generally, for a CP-mixed coupling  $\bar{\tau}(\cos\Delta + i\gamma_5 \sin\Delta)\tau$  the transverse block rotates,  $C_{nn} = C_{rr} = \cos 2\Delta$ ,  $C_{nr} = -C_{rn} =$

$\sin 2\Delta$ ,  $C_{kk} = -1$ , so that  $C^T C = \mathbb{K}$  and the state remains maximally entangled for any  $\Delta$ : the Horodecki parameter [13]—the sum  $m_{12}$  of the two largest eigenvalues of  $C^T C$ —equals 2, the concurrence equals 1, and the maximal CHSH score is  $2\sqrt{m_{12}} = 2\sqrt{2}$ , the Tsirelson bound. CP violation rotates the correlations but does not dilute them; entanglement and CP structure are measured simultaneously from the same matrix.

*Extraction from data.*—In  $\tau \rightarrow \pi\nu$  the pion is a perfect spin analyser ( $\alpha_\pi = 1$ ) [22]. With  $\hat{h}^\pm$  the  $\pi^\pm$  direction in the  $\tau^\pm$  rest frame, the joint angular density is

$$p(\hat{h}^+, \hat{h}^- | C) = \frac{1}{(4\pi)^2} (1 - \hat{h}^{+\top} C \hat{h}^-), \quad (3)$$

where the sign reflects  $\alpha_{\pi^+} \alpha_{\pi^-} = -1$ . The moment estimator is  $\hat{C}_{ij} = -9 \langle \hat{h}_i^+ \hat{h}_j^- \rangle$ , the factor 9 following from  $\langle (\hat{h}_i)^2 \rangle = \frac{1}{3}$ . The  $\pi^+ \pi^-$  acoplanarity angle  $\phi$  (the azimuthal opening of the two decay planes about  $\hat{k}$ ) inherits the modulation

$$\frac{d\Gamma}{d\phi} \propto 1 - \frac{\pi^2}{16} \cos(\phi - 2\Delta), \quad (4)$$

where  $\pi^2/16 \simeq 0.617$  combines  $C_{nn} = C_{rr} = 1$  with  $\langle \sin^2 \theta \rangle = (\pi/4)^2$  from each polar average [23, 24].

*Estimators and their sampling properties.*—A reliable nonlocality claim at small  $N$  requires care: the eigenvalue construction makes  $m_{12}$  a *noise-biased* statistic. Under the no-correlation null,  $\hat{C} \rightarrow (3/\sqrt{N})G$  with  $G_{ij}$  i.i.d. standard normal, so  $\hat{m}_{12}$  is  $9/N$  times the sum of the two largest eigenvalues of a  $3 \times 3$  Wishart matrix with three degrees of freedom:

$$E[\hat{m}_{12} | C = 0] = \frac{9 \langle \lambda_1 + \lambda_2 \rangle}{N} = \frac{78.5}{N}. \quad (5)$$

For  $N = 50$  this “noise floor” is 1.6—above the Bell threshold  $m_{12} = 1$ —so significance counting based on  $\hat{m}_{12}$  alone is meaningless at small  $N$ . We therefore base all hypothesis tests on statistics *linear* in  $C$  (hence unbiased) or on exactly calibrated likelihood ratios:

(1) *CHSH score with fixed axes.* Choosing the measurement axes a priori from the SM prediction ( $\hat{a} = \hat{n}$ ,  $\hat{a}' = \hat{r}$ ,  $\hat{b}, \hat{b}' = (\hat{n} \pm \hat{r})/\sqrt{2}$ ), the CHSH combination is  $\langle S \rangle = \sqrt{2}(C_{nn} + C_{rr})$ , estimated by the per-event quantity  $s = -9\sqrt{2}(\hat{h}_n^+ \hat{h}_n^- + \hat{h}_r^+ \hat{h}_r^-)$  with  $E[s] = 2\sqrt{2}$  and  $\text{Var}[s] = 28$  under the SM (exactly), and  $\text{Var}[s] = 36$  under the null. Local hidden-variable theories require  $S \leq 2$  [3].

(2) *Concurrence witness.* The fidelity to the Bell state,  $F = \langle \psi^+ | \rho | \psi^+ \rangle = (1 + C_{nn} + C_{rr} - C_{kk})/4$ , bounds the concurrence from below,  $\mathcal{C} \geq \mathcal{W} \equiv 2F - 1$  [14]; any separable state has  $\mathcal{W} \leq 0$  while the SM gives  $\mathcal{W} = 1$ . The per-event estimator has  $\text{Var}[w] = 9/2$  under the SM (exactly), so separability is rejected at  $5\sigma$  with only  $N \gtrsim 113$  events.

(3) *Optimal no-correlation test.* Against the hypothesis that the two decays are uncorrelated ( $p_0 = (4\pi)^{-2}$ ), the Neyman–Pearson optimal statistic is the per-event log-likelihood ratio  $t = \ln(1 - \hat{h}^{+\top} C_{\text{SM}} \hat{h}^-)$ , for which we find the closed-form moments

$$\begin{aligned} E_{\text{SM}}[t] &= \ln 2 - \frac{1}{2}, & \text{Var}_{\text{SM}}[t] &= \frac{1}{4}, \\ E_0[t] &= \ln 2 - 1, & \text{Var}_0[t] &= 1. \end{aligned} \quad (6)$$

The expected significance for rejecting “no correlation” is therefore  $Z = \sqrt{N}(E_{\text{SM}} - E_0)/\sqrt{\text{Var}_0} = \sqrt{N}/2$  exactly. The test is calibrated without any Gaussian assumption by *permutation*: re-pairing  $\tau^+$  and  $\tau^-$  hemispheres across events realises the null exactly in data, preserving all single-tau distributions and detector effects while destroying the correlations.

*Causal-speed distribution.*—From the reconstructed decay four-positions we form  $\Delta s^2 = c^2 \Delta t^2 - \Delta r^2$  (invariant) and the minimum signal speed required to connect the decays,  $v_{\text{sig}} = \Delta r/(c \Delta t)$ , evaluated in the Higgs rest frame. There the taus are exactly back-to-back with equal speed  $\beta$ , and decay lengths  $L_\pm$  are i.i.d. exponential. Since  $(L_+ - L_-)/(L_+ + L_-)$  is exactly uniform on  $[-1, 1]$  for exponential variables,

$$v_{\text{sig}} = \beta \frac{L_+ + L_-}{|L_+ - L_-|} \implies P(v_{\text{sig}} > v) = \frac{\beta}{v}, \quad v \geq \beta, \quad (7)$$

an exact Pareto law: half of all events require superluminal connections faster than  $2c$ , and a fraction  $1/v$  faster than  $vc$ . Under a finite-speed causal model [7, 8] in which correlations are carried by an influence propagating at  $v_\psi$  in the Higgs frame, all events with  $v_{\text{sig}} > v_\psi$  must be uncorrelated. Combining Eq. (7) with  $Z = \sqrt{N}/2$ , the median 95% C.L. exclusion reach is

$$v_\psi^{95} = \frac{N\beta}{4 z_{95}^2} \simeq \frac{N}{15.4}, \quad (8)$$

*linear* in the event count  $N$ .

*Simulation and reconstruction.*—We analyse simulated  $e^+e^- \rightarrow ZH \rightarrow \mu^+\mu^-\tau^+\tau^-$  events at  $\sqrt{s} = 240$  GeV with both taus decaying to  $\pi\nu$ . From 1003 generated  $\mu^+\mu^-\tau^+\tau^-$  events, 90 satisfy the  $\pi \times \pi$  requirement and are fully reconstructed; this corresponds to  $\sim 15 \text{ ab}^{-1}$  in this single channel, and we use Eq. (8) to project to realistic multi-channel datasets below. The Higgs four-momentum is  $p_H = p_{\text{beam}} - p_{\mu^+} - p_{\mu^-}$ , exploiting the known beam energy. The tau direction and decay vertex follow from the impact-parameter method [25]: the pion track and the production vertex define a plane containing the tau momentum; the opening angle  $\alpha$  fixes the decay length  $L = d/\sin \alpha$  and the  $\tau$  mass constraint fixes the momentum, with the two-fold ambiguity resolved by the missing-momentum constraint  $p_H - p_{\pi^+} - p_{\pi^-}$ . Tau energies are then set by the Higgs-frame constraint  $E_\tau = m_H/2$ . A vertex-quality requirement of  $3\sigma$  impact-parameter significance on both taus

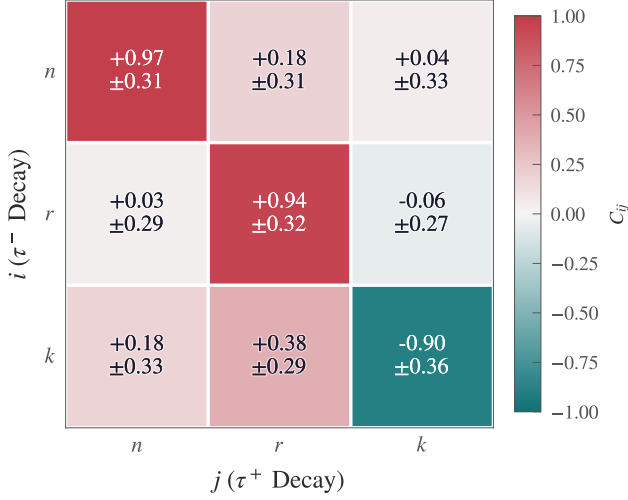


FIG. 1. Reconstructed spin correlation matrix  $C_{ij}$  in the  $\{n, r, k\}$  basis from 90  $\tau \rightarrow \pi\nu$  pairs. Entries show the central value  $\pm$  bootstrap uncertainty. The exact SM prediction, Eq. (2), is  $C = \text{diag}(+1, +1, -1)$  for any  $\beta$ .

retains 51 events for vertex-dependent observables; the spin observables use all 90. Detector response is modelled by Gaussian smearing of ILD-like track parameters ( $\sigma_{d_0} = \sqrt{5^2 + (10/p \sin^{3/2} \theta)^2} \mu\text{m}$ ,  $\sigma_{\theta, \phi} = 0.1 \text{ mrad}$ ,  $\sigma_{1/p_T} \simeq 2 \times 10^{-5} \text{ GeV}^{-1}$ ) [28]; the typical tau impact parameter at  $\gamma \approx 35$  is comparable to  $\sigma_{d_0}$ , so smearing broadens  $v_{\text{sig}}$  by a measured 27% fractional resolution while leaving the  $C_{ij}$  extraction essentially intact.

*Spin correlations.*—Figure 1 shows the reconstructed correlation matrix, consistent with Eq. (2): the diagonal is  $(+0.97, +0.94, -0.90) \pm 0.3$ , off-diagonals consistent with zero. The unbiased observables give

$$S = 2.71 \pm 0.57, \quad \mathcal{W} = 0.91 \pm 0.24, \quad (9)$$

both consistent with the SM expectations  $2\sqrt{2}$  and 1, and with truth-level validation ( $S = 2.81 \pm 0.58$ ,  $\mathcal{W} = 0.91 \pm 0.24$ ). Separability ( $\mathcal{W} \leq 0$ ) is rejected at  $3.8\sigma$ : the tau pairs are certifiably entangled. The raw Horodecki parameter,  $m_{12} = 2.34 \pm 1.00$ , illustrates Eq. (5): its noise floor at  $N = 90$  is 0.87, and the bootstrap-bias-corrected value is 1.26; we therefore do not base significance claims on it. The optimal LR test rejects “no correlation” with permutation  $p < 10^{-4}$  ( $Z = 4.3$ , against the ideal expectation  $\sqrt{90}/2 = 4.7$ ). The truth-level acoplanarity modulation,  $B = -0.67 \pm 0.13$ , confirms the exact prediction  $-\pi^2/16 = -0.617$  of Eq. (4) (a frequently quoted value of  $-0.5$  is not the SM expectation for this channel). With full detector smearing the central values shift by less than one standard deviation ( $S = 2.34 \pm 0.61$ ,  $\mathcal{W} = 0.79 \pm 0.27$ , LR  $Z = 3.3$ ).

*Spacetime-resolved entanglement.*—Figure 2 shows  $m_{12}$  (bias-corrected) and the witness  $\mathcal{W}$  versus the signed

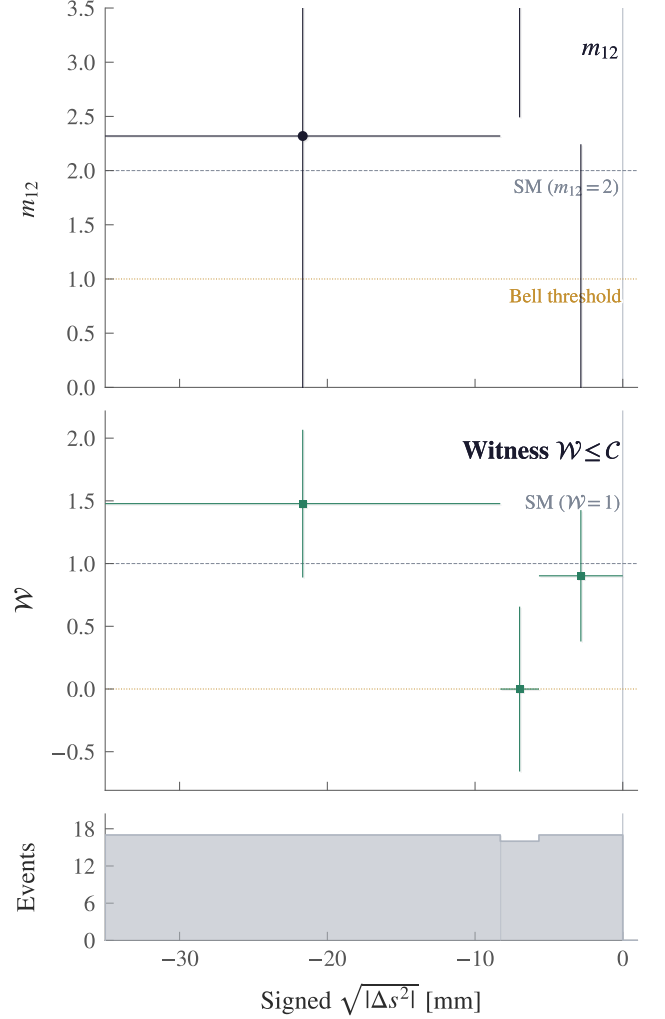


FIG. 2. Bias-corrected  $m_{12}$  (top) and concurrence witness  $\mathcal{W}$  (middle) versus the signed spacetime interval between the two tau decay vertices; negative values are spacelike. Bottom: event counts. Dashed lines mark the SM predictions and the Bell ( $m_{12} = 1$ ) and separability ( $\mathcal{W} = 0$ ) thresholds.

interval  $\text{sgn}(\Delta s^2) \sqrt{|\Delta s^2|}$ . All 51 vertex-selected events are spacelike separated, a consequence of the back-to-back topology, the ultrarelativistic boost, and the vertex-quality requirement (timelike configurations require one near-prompt decay, which fails the impact-parameter cut). Entanglement persists across the spacelike region with no significant dependence on the interval: the correlations connect events that no subluminal signal could connect.

*Finite-speed exclusion.*—For each hypothesis  $v_\psi$  we select the events with  $v_{\text{sig}} > v_\psi$  and apply the permutation-calibrated LR test. Figure 3 shows the result: the no-correlation hypothesis is rejected at 95% C.L. for  $v_\psi \lesssim 3c$ , in agreement with the analytic expectation  $v_\psi^{95} = 3.3c$  from Eq. (8) with  $N = 51$ . The measured  $v_{\text{sig}}$  distribu-

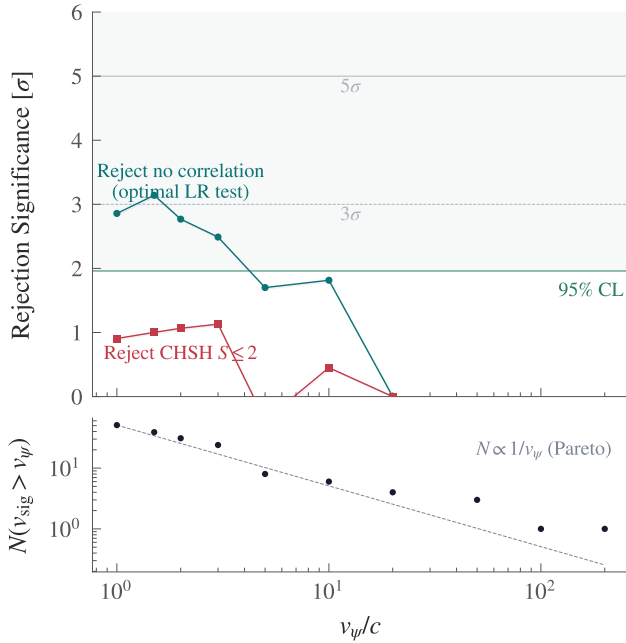


FIG. 3. Significance for rejecting the finite-speed causal model versus the hypothesised speed  $v_\psi$ . Circles: optimal LR test against “no correlation” (permutation-calibrated); squares: CHSH  $S \leq 2$  rejection. Horizontal lines mark 95% C.L.,  $3\sigma$  and  $5\sigma$ . Bottom: surviving event counts, following  $P(v_{\text{sig}} > v_\psi) = \beta/v_\psi$ .

tion follows the Pareto law of Eq. (7) (observed median 2.6, predicted  $2\beta \simeq 2$ , with the difference attributable to resolution and selection).

*Projections.*—Equation (8), validated above on simulation, converts event counts directly into physics reach. At the full ILC program ( $2 \text{ ab}^{-1}$  at  $\sqrt{s} = 250 \text{ GeV}$  [27]), about  $3.0 \times 10^4$   $H \rightarrow \tau^+\tau^-$  decays are produced. Including  $Z \rightarrow q\bar{q}$  and  $Z \rightarrow ee, \mu^+\mu^-$  (76% of  $Z$  decays) and the  $\pi/\rho$  tau modes analysed with polarimeter vectors [24, 25],  $N \simeq 3 \times 10^3$  reconstructed pairs are expected, giving  $v_\psi^{95} \approx 2 \times 10^2 c$  and a  $5\sigma$  Bell-inequality violation ( $N \gtrsim 10^3$  suffices for  $S > 2$  at  $5\sigma$ , and  $N \gtrsim 10^2$  for entanglement certification). Using all hadronic tau modes raises  $N$  towards  $10^4$  and the reach towards  $10^3 c$ ; realistic reconstruction dilution (we measure  $Z_{\text{obs}}/Z_{\text{ideal}} \simeq 0.7\text{--}0.9$  per event with full smearing) reduces these figures by a factor of about two.

*Discussion.*—Dedicated photon experiments bound the speed of a hypothetical “spooky action” to  $\gtrsim 10^4 c$  for particular preferred frames [7], far above the reach derived here. The measurement we propose is nevertheless qualitatively new in three respects. First, the entangled system consists of *massive fermions* produced by the *Yukawa interaction*—a sector in which nonlocality has never been tested; given the weak sector’s record of broken symmetries, this is a test worth performing,

not assuming. Second, the spacelike separation is not engineered but occurs naturally at the micrometre-to-millimetre scale, probing causal structure at microscopic distances and at a relativistic boost ( $\gamma \approx 35$ ) inaccessible to optical experiments. Third, the same dataset simultaneously determines the CP structure of the  $H\tau\tau$  coupling through the acoplanarity phase  $2\Delta$  in Eq. (4): nonlocality and CP violation are measured from one correlation matrix. As in all decay-based tests, the “measurement settings” are not free choices, so the freedom-of-choice loophole remains open [26]; the measurement excludes the class of finite-speed causal models defined above rather than all local theories.

In summary, we have presented the complete statistical framework for a spacetime-resolved entanglement measurement in  $H \rightarrow \tau^+\tau^-$  at an  $e^+e^-$  Higgs factory: exact SM expectations [Eqs. (2), (4)], unbiased estimators with closed-form variances, an optimal permutation-calibrated test [Eq. (6)], and an analytic, simulation-validated sensitivity law [Eqs. (7), (8)]. The measurement certifies entanglement between spacelike-separated, Yukawa-produced massive fermions and excludes finite-speed causal models up to speeds growing linearly with the dataset—a unique capability of future Higgs factories.

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